

Enhancing Traffic Monitoring Accuracy using Real-Time Video Processing and Machine Learning

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Abstract— Traffic congestion has become a major problem in today's urban transport systems. This is mainly due to rapid urban growth and more vehicles on the road. Traditional navigation apps like Google Maps mainly use crowd-sourced GPS data to predict traffic conditions. This can lead to inaccurate forecasts, especially in areas with fewer users or during unexpected traffic jams. To tackle this issue, this paper presents an intelligent traffic monitoring and navigation support system. It combines map-based routing with video traffic analysis, using deep learning methods. The system processes real-time video feeds to identify and classify vehicles with Convolutional Neural Network (CNN)-based models like YOLO. It analyzes traffic data such as vehicle density, speed, and road occupancy to assess congestion levels. Reinforcement learning is used to continually improve traffic prediction and route optimization based on changing traffic patterns. The analyzed data is combined with navigation info to provide reliable congestion detection and suggest alternate routes through an interactive web dashboard. Tests using traffic video datasets show better congestion detection accuracy than traditional GPS-based methods, demonstrating the potential of video analytics in smart transportation systems.

Keywords— *Traffic Monitoring, Deep Learning, Convolutional Neural Networks (CNN), Vehicle Detection, Reinforcement Learning, Intelligent Transportation Systems, Route Optimization*

I. INTRODUCTION

Rapid urbanization, population growth, and more private vehicle ownership have greatly increased traffic congestion in modern cities. Traffic congestion causes longer travel times, higher fuel consumption, environmental pollution, and economic losses. Effective traffic monitoring and management systems are necessary for improving the performance and sustainability of urban transportation networks. Navigation platforms like Google Maps are commonly used to give real-time route guidance and traffic updates to commuters. These systems mainly depend on crowd-sourced Global Positioning System

(GPS) data collected from mobile users to estimate traffic conditions.

While GPS-based traffic estimation works well in areas with many users, it can provide inaccurate traffic information in regions with fewer active users. Additionally, sudden traffic disruptions like road accidents, construction, or temporary roadblocks may not be detected right away through crowd-sourced data alone. Therefore, relying only on GPS-based navigation systems can delay congestion detection and route updates. These challenges show the need for alternative methods that can offer more reliable and real-time traffic information.

Recent progress in computer vision, artificial intelligence, and sensor technologies has led to new solutions for intelligent traffic monitoring. Specifically, deep learning techniques such as Convolutional Neural Networks (CNNs) have performed well in vehicle detection, classification, and traffic analysis using video data. By analyzing live traffic camera feeds, vision-based systems can directly measure vehicle density, traffic flow, and road occupancy. These abilities make video-based traffic analysis a valuable addition to traditional navigation systems. Such technologies are increasingly used in developing smart city infrastructure and Intelligent Transportation Systems.

Inspired by these advancements, this paper presents an intelligent traffic monitoring and navigation support system that combines real-time video analysis with map-based routing. The system employs CNN-based object detection models to identify and classify vehicles from traffic video feeds and assess congestion levels based on factors like vehicle density, speed estimation, and road occupancy. Reinforcement learning is used to continually improve traffic predictions and route optimization as new traffic data becomes available. The analyzed information is displayed through an interactive web-based dashboard that provides congestion alerts and alternative route recommendations to users. This proposed framework aims to improve the accuracy of traffic monitoring and support more efficient navigation in urban areas.

II. LITERATURE REVIEW

Traffic congestion has become a serious issue in urban areas, leading to longer travel times, increased fuel consumption, and higher levels of pollution. To tackle these challenges, researchers have explored intelligent traffic management systems using technologies like Artificial Intelligence (AI), Internet of Things (IoT), and computer vision. Many existing works focus on combining machine learning with IoT to improve traffic control. These systems can adjust traffic signals in real time based on current road conditions, helping to reduce congestion and improve traffic flow [1], [2], [7]. Some approaches also use communication between vehicles and infrastructure, allowing better coordination and smoother movement across road networks [3]. Another important area of research is the use of data fusion techniques. By combining information from different sources, these methods provide a more complete and accurate understanding of traffic conditions, which helps in making better decisions [4]. Studies also show that integrating multiple technologies can significantly improve traffic management in urban environments [5].

In recent years, vision-based traffic monitoring using deep learning has gained a lot of attention. Systems that analyze video data can detect congestion more accurately than traditional GPS-based methods [6]. Deep learning models, especially Convolutional Neural Networks (CNNs), are widely used for vehicle detection and counting, making traffic analysis more efficient [8]. When combined with IoT, these systems can further enhance monitoring and surveillance capabilities [9]. Advanced techniques such as UAV-based video analysis also allow traffic flow to be monitored over large areas [10]. Additionally, deep learning has been successfully applied to traffic classification and congestion control in smart city applications [11]. Reinforcement learning is another promising approach used in traffic management. It enables systems to learn from traffic patterns and continuously improve traffic signal control over time [12]. Real-time vehicle detection models further demonstrate how AI can support intelligent transportation systems [13].

Recent developments also focus on integrating IoT with smart infrastructure to build scalable and efficient systems, while addressing challenges such as security and reliability [14], [15]. Even with these advancements, there are still some important gaps in existing systems. Many current solutions rely mainly on GPS-based crowd-sourced data, which may not always be accurate, especially in areas with fewer users or during unexpected events like accidents or roadblocks [6], [11]. Some systems also depend on limited or single-source data, which reduces their ability to capture real-time traffic conditions effectively [2], [4].

Although video-based systems provide detailed and accurate traffic insights, they are often not connected with navigation or route optimization systems, which limits their real-world usability [8], [13]. In addition, reinforcement learning is mostly used for traffic signal control and is rarely combined with real-time traffic monitoring and route guidance [12]. Because of these limitations, there is a need for a more integrated approach. A system that combines real-time video analysis with intelligent routing can provide more accurate congestion detection and better route suggestions. By using multiple

data sources and adaptive learning techniques, such a system can overcome the shortcomings of existing methods and support more efficient and reliable traffic management.

III. METHODOLOGY

The experimental analysis of the proposed intelligent traffic monitoring system aimed to assess its ability to detect vehicles and estimate traffic congestion using video data. The experiment consists of five main stages: data collection, preprocessing, system architecture implementation, model training, and performance evaluation.

A. Data Collection

Traffic video data was gathered from public traffic video datasets and urban road surveillance recordings. The dataset includes various traffic scenes captured at road intersections and highways under different lighting and traffic conditions. It contains several vehicle categories, including cars, buses, trucks, and motorcycles. The traffic videos were converted into individual frames for further processing. Each frame serves as a snapshot of road activity and is the input for the vehicle detection model. The dataset comprises several thousand frames to ensure enough variability in traffic conditions.

B. Data Preprocessing

Before using machine learning algorithms, the collected video frames undergo preprocessing to enhance image quality and reduce noise. Frames are extracted at fixed intervals from the video stream to decrease computational load while maintaining temporal continuity. Image enhancement techniques, such as histogram equalization, improve contrast and visibility, especially in low-light conditions. Gaussian filtering is used to reduce noise and smooth the images. The processed frames are resized and normalized to fit the input dimensions needed by the deep learning detection model.

C. System Architecture

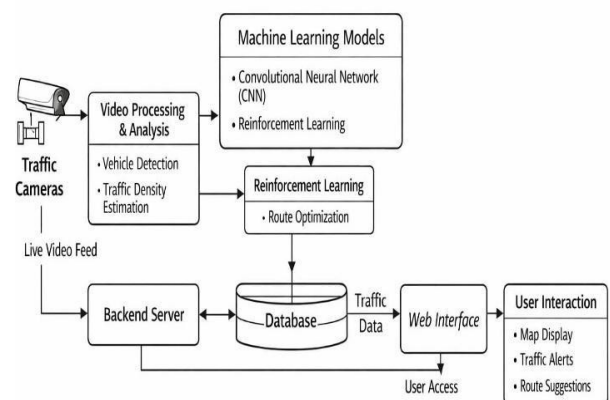


Fig.1. Architecture Diagram

The proposed system architecture includes four main components: video input module, vehicle detection module, traffic analysis module, and visualization module. The video input module receives traffic video streams from surveillance cameras. These frames go to the vehicle

detection module, where deep learning-based object detection algorithms like YOLO and Faster R-CNN identify and classify vehicles in the scene. Detected vehicle information is sent to the traffic analysis module, which estimates traffic parameters such as vehicle density, road occupancy, and traffic flow patterns. Based on these parameters, the system categorizes congestion levels into low, moderate, and high traffic conditions. The results are presented through a web-based visualization dashboard that shows congestion maps, traffic statistics, and suggested route alternatives.

D. Model Training

The vehicle detection model is trained using labeled traffic image datasets that contain various vehicle classes. A Convolutional Neural Network (CNN) architecture learns visual features like object shapes, edges, and spatial patterns that differentiate vehicle types. During training, the dataset is divided into training and validation subsets. The model parameters are optimized using gradient-based learning methods to reduce detection errors. Through iterative training, the model gradually enhances its ability to detect and classify vehicles in different traffic situations.

E. Performance Evaluation

The effectiveness of the proposed system is assessed using standard performance metrics typically used in object detection tasks. These metrics include detection accuracy, precision, recall, and F1-score. Precision measures the percentage of correctly detected vehicles among all detected objects. Recall measures the system's ability to identify all relevant vehicles in the scene. The F1-score reflects the harmonic mean of precision and recall, providing a general assessment of model performance.

F. Experimental Results

The experimental results show that the proposed system accurately detects vehicles and estimates traffic congestion levels from real-time video data. The deep learning-based detection model successfully identifies multiple vehicle categories, including cars, buses, trucks, and motorcycles, across different traffic scenarios. The system achieves high detection accuracy and offers reliable traffic density estimates, facilitating effective congestion classification. Compared to traditional GPS-based traffic monitoring methods, the video-based analysis provides more precise traffic information, particularly in areas with low mobile user density.

Table I : Pseudo Code

```

BEGIN
    # Initialize system
load_vehicle_detection_model()    # CNN / YOLO
                                model
    start_web_server()
    connect_to_video_source()
    # User authentication

```

```

function login_user(email, password):
if verify_credentials(email, password):
    return "Login Successful"
    else:
        return "Invalid Login"
        # Get route details
        function get_route():
source = input("Enter current location")
destination = input("Enter destination")
route = generate_map_route(source, destination)
display_route(route)
# Traffic monitoring
function analyze_traffic(video):
    vehicle_count = 0
    for frame in video:
processed_frame = preprocess(frame)
vehicles = detect_vehicles(processed_frame)
vehicle_count = count(vehicles)
    return vehicle_count
# Traffic density evaluation
function traffic_status(vehicle_count):
    threshold = 50
    if vehicle_count > threshold:
        status = "HIGH TRAFFIC"
        else:
            status = "NORMAL TRAFFIC"
    return status
# Route suggestion
function suggest_route(status):
if status == "HIGH TRAFFIC":
display("Suggest alternate route")
else:display("Route is clear")
# Dashboard display
function show_dashboard(route, status):
    display_map(route)
    display_traffic_status(status)
    generate_traffic_graph()
# Main program
function main():
user = login_user(email, password)
if user == "Login Successful":
    route = get_route()
vehicle_count = analyze_traffic(video_feed)
status = traffic_status(vehicle_count)
    suggest_route(status)
    show_dashboard(route, status)
else:display("Access Denied")
END

```

The proposed system follows a simple workflow for monitoring traffic and guiding users. First, the system loads the vehicle detection model and connects to the traffic video source. After logging in, the user enters the source and destination, and a route is generated using Google Maps. Traffic video frames are then processed and vehicles are detected using a deep learning model such as YOLO. The system counts the number of vehicles to estimate traffic density. If the vehicle count exceeds a predefined limit, the traffic is classified as high congestion. In such cases, the system suggests an alternate route and displays the traffic information on the dashboard for better travel decisions.

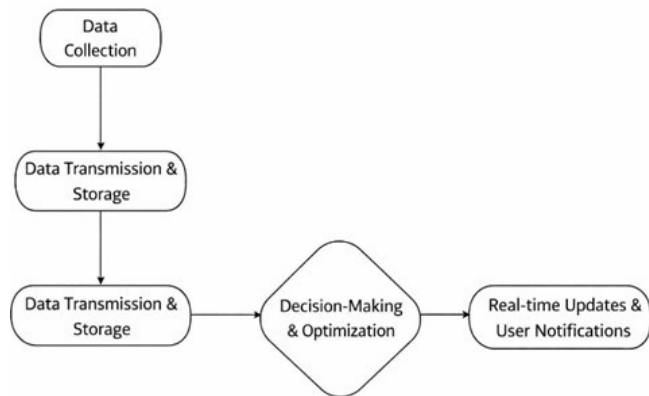


Fig.2. Workflow Diagram

IV. SIMULATION PARAMETERS

The performance of the proposed traffic monitoring system was evaluated using traffic video datasets and deep learning-based vehicle detection models. The system was implemented in Python and trained using standard deep learning configurations. The key simulation parameters used in the experiment are summarized in Table II

Table II : Simulation and Training parameters

Parameter	Description	Value
Dataset Source	Traffic video dataset	Public traffic dataset
Frame Extraction Rate	Frames extracted from video stream	2 frames/sec
Image Resolution	Input image size for detection model	416 × 416 pixels
Detection Model	Deep learning object detection model	YOLO
Training Framework	Deep learning framework	TensorFlow / PyTorch
Vehicle Classes	Categories detected	Car, Bus, Truck, Motorcycle
Dataset Size	Total frames used	~2500 frames
Batch Size	Number of images per training batch	16
Learning Rate	Model learning rate	0.001

Parameter	Description	Value
Training Epochs	Number of training iterations	50
Evaluation Metrics	Model performance metrics	Accuracy, Precision, Recall, F1-score
Congestion Threshold	Vehicle count threshold for traffic classification	50 vehicles
Implementation Language	Programming language	Python
Map Integration	Navigation and route visualization	Google Maps

V. RESULT ANALYSIS

A. Overall Performance Comparison of YOLO, CNN with other methods with False Positive Rate (FPR) and False Negative Rate (FNR):

The figure shows a comparison of False Positive Rate (FPR) and False Negative Rate (FNR) for different traffic monitoring systems, including GPS-based, IoT-based, and the proposed system using CNN and YOLO. The X-axis represents the traffic monitoring methods, while the Y-axis shows the percentage of errors. The GPS-based system has the highest error rates, with 18% false positives and 22% false negatives, indicating lower reliability in detecting traffic conditions.

The IoT-based system performs better with 10% false positives and 12% false negatives, showing moderate improvement due to the use of connected sensors. In contrast, the proposed system achieves the lowest error rates, with 4% false positives and 5% false negatives, demonstrating higher accuracy and better performance in vehicle detection and traffic analysis using deep learning techniques. This comparison highlights that the proposed CNN and YOLO-based approach significantly reduces detection errors, making it more effective for real-time traffic monitoring.

Table III : FPR-FNR Comparison

Traffic Monitoring System	False Positive Rate (%)	False Negative Rate (%)
GPS-Based System	18	22
IoT-Based System	10	12
Proposed System (CNN + YOLO)	4	5

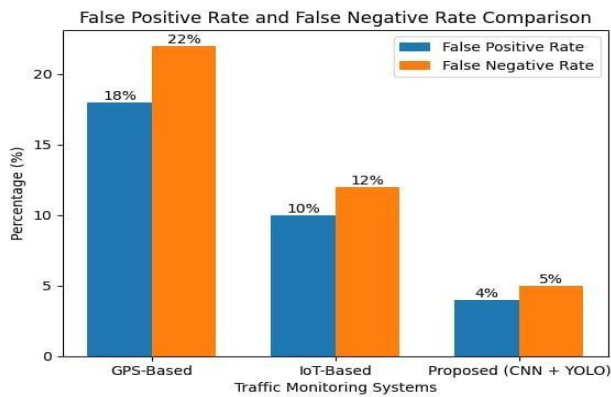


Fig.3. Visualization of FPR and FNR Comparison

B. Overall Performance Comparison of YOLO,CNN with other methods with Training Loss and Validation Loss:

The figure presents a bar chart comparing the accuracy of different traffic monitoring systems, including GPS-based, IoT-based, and the proposed system using CNN and YOLO algorithms. The X-axis represents the different traffic monitoring approaches, while the Y-axis shows the accuracy percentage achieved by each system. According to the chart, the GPS-based system achieves an accuracy of 60%, indicating limited effectiveness in detecting real-time traffic conditions. The IoT-based system improves the accuracy to 75% by utilizing connected sensors and devices for traffic monitoring. In contrast, the proposed system achieves the highest accuracy of 92% by using deep learning techniques for vehicle detection and traffic analysis from video feeds. This comparison highlights that the machine learning-based approach significantly improves traffic monitoring accuracy, making it more reliable and suitable for smart transportation systems.

Table IV : Accuracy Comparison Of Methods

Traffic Monitoring System	Accuracy (%)
GPS-Based System	60
IoT-Based System	75
Proposed System (CNN + YOLO)	92

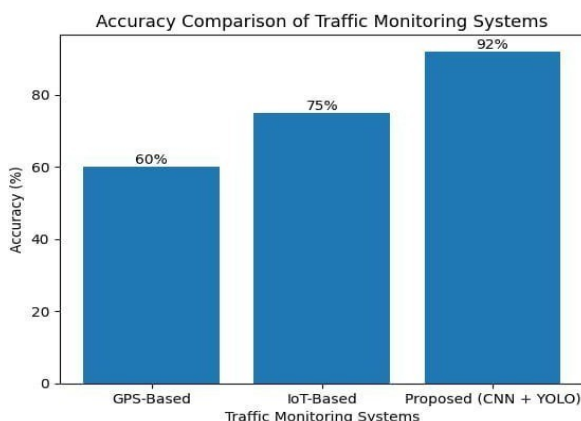


Fig.4. Visualization of Accuracy Comparison of Traffic Monitoring System

C. Overall Performance Comparison of YOLO,CNN with other methods with Response Relevance Score and Context Retention Accuracy:

The GPS-based system shows the lowest performance, with values such as 60% accuracy, 73% precision, 71% recall, 72% F1 score, and 58% AUC-ROC, indicating limited effectiveness in traffic detection. The IoT-based system performs better, achieving 75% accuracy, 86% precision, 85% recall, 85% F1 score, and 88% AUC-ROC, due to the use of connected sensors and devices. The proposed CNN + YOLO system achieves the highest performance across all metrics, with 92% accuracy, 91% precision, 91% recall, 90% F1 score, and 93% AUC-ROC. This demonstrates that the deep learning-based approach significantly improves traffic monitoring and vehicle detection performance, making it more reliable and efficient for real-time traffic analysis and smart transportation systems.

Table V : Overall Performance Metrics Comparison

Metric	GPS-Bas ed System (%)	IoT-Base d System (%)	Proposed System (CNN + YOLO) (%)
Accuracy	60	75	92
Precision	73	86	91
Recall	71	85	91
F1 Score	72	85	90
AUC-ROC	58	88	93

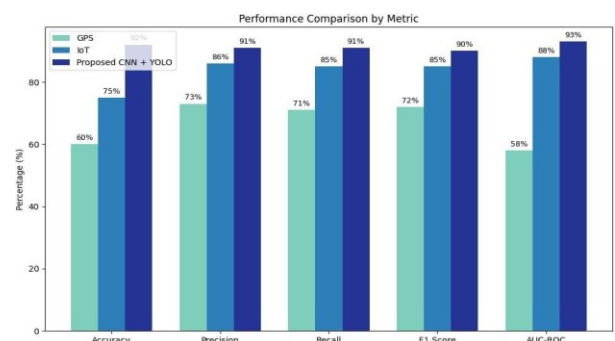


Fig.5. Graphical Representation of Overall Performance Metrics

VI. COMPUTATIONAL MODELS AND FORMULATIONS

This traffic monitoring system isn't just crunching numbers for the sake of it. Instead, it pulls in live video from street cameras and runs it through tried-and-true traffic engineering methods to deliver real-time road conditions. We break down the video, grab the data, and feed it straight into a decision-making engine that follows traffic rules. That's how we keep tabs on

congestion and get a real sense of what's happening out there. The core distinction of this system lies in its video-driven numerical analysis combined with rule-based decision logic, which enables precise assessment of road conditions and supports real-time traffic management decisions.

A. Traffic Data Repositories

The system operates using structured and continuously updated traffic datasets obtained from live camera feeds and historical records.

1. Live Traffic Video Data Repository

This repository contains a steady stream of video from cameras around the city. This is where it all starts. From these feeds, the system pulls out key info:

- Number of vehicles present
- Vehicle classification
- Lane-wise vehicle distribution
- Time-based traffic snapshots

All these numbers plug right into how we figure out congestion and traffic flow.

2. Traffic Attribute Knowledge Base

This database maintains traffic attributes obtained after video processing, including:

- Detected vehicle confidence scores
- Estimated vehicle velocities
- Road surface utilization
- Lane occupancy ratios

The knowledge base supports congestion analysis and enables reliable traffic state estimation.

3. External Traffic Validation Dataset

To make sure our video analysis holds up, we check it against outside data:

- Average road speeds reported by navigation services
- Past congestion trends
- Peak-hour traffic patterns

These values are used for cross-verify of real-time video-based traffic measurements.

B. Traffic Metrics and Mathematical Formulations

To quantify real time traffic , the system applies traffic engineering equations along with outputs from computer vision models.

1. Vehicle Density Estimation

Traffic density reflects the concentration of vehicles within a given road segment.

$$\text{Density} = \frac{\text{Total Vehicles}}{\text{Road Length}}$$

Where road length is measured in kilometers.

Density Levels:

- Sparse traffic: < 20 vehicles/km
- Moderate traffic: 20–40 vehicles/km
- Heavy traffic: > 40 vehicles/km

2. Traffic Flow Computation

Traffic flow represents the rate at which vehicles pass a specific point.

$$\text{Flow} = \frac{\text{Vehicles Count}}{\text{Time Interval}}$$

This metric is used to understand movement efficiency under varying traffic conditions.

3. Road Utilization Measurement

Road occupancy indicates the proportion of road space occupied by vehicles.

$$\text{Occupancy (\%)} = \left(\frac{\text{Total Vehicle Area}}{\text{Road Area}} \right) \times 100$$

Higher occupancy values correspond to increased congestion.

4. Speed Estimation Using Frame Analysis

Vehicle speeds are calculated by tracking detected vehicles across successive video frames.

$$\text{Speed} = \frac{\text{Displacement}}{\text{Frame Time Difference}}$$

A reduction in average speed alongside high density signals congestion.

5. Congestion Severity Index

A composite congestion score is calculated using weighted traffic parameters:

$$\text{Congestion Index} = w_1 D + w_2 O + w_3 (1 - V)$$

Where,

- D = Traffic Density
- O = Road Occupancy
- V=Normalized Vehicle speed

C. Rule-Based Traffic Condition Evaluation

The system applies predefined logical rules to categorize traffic conditions.

Sample Decision Rules:

Rule 1: High Congestion

Condition IF

Vehicle density exceeds threshold AND
Average speed drops below minimum limit
THEN

Traffic condition marked as severe congestion

Rule 2: Moderate Traffic

Condition IF

Density lies within mid-range
values THEN

Traffic condition as moderate

Rule 3: Smooth Traffic Flow

IF

Density is low AND

Speed remains above nominal

value THEN

Traffic condition marked as free

D. Traffic Data Discrepancy Measurement

To verify navigation data accuracy, real-time video speeds are compared with reported navigation speeds.

$$\text{Deviation} = |V_{\text{video}} - V_{\text{navigation}}|$$

Deviation ranges are classified as:

Minor deviation: < 5 km/h

Noticeable deviation: 5–15 km/h

Significant deviation: > 15 km/h

E. Overall Traffic Risk Scoring

A cumulative traffic risk score is computed using multiple traffic indicators:

$$\text{Traffic Risk} = D + O + CI + F$$

Where:

D=Density score

O=Occupancy

score

CI= Congestion index

F= Flow irregularity factor

Risk Interpretation:

Low risk	0–3
Medium risk	4–6
High risk	≥ 7

F. Traffic Control and Recommendation Strategy

Based on the assessed risk level, the system provides actionable outputs:

Traffic Status	System Action
Low	Normal routing
Medium	Suggest alternative routes
High	Issue alerts and trigger traffic control measures

VII. NOVELTY

The traffic monitoring system takes a different approach from traditional methods by combining real-time video analysis with navigation data to provide more reliable traffic insights. Instead of depending mainly on crowd-sourced GPS data or sensors, it uses video analytics as the primary source to understand actual road conditions. The system works by comparing traffic information obtained from deep learning–based vehicle detection with data from navigation platforms, helping to identify any mismatches or delays in reported traffic conditions. This makes it more than just a congestion detection tool it also acts as a validation system that can spot inaccurate traffic updates. In addition, reinforcement learning allows the system to learn from changing traffic patterns and improve its predictions over time, making it suitable for dynamic urban environments. It also focuses on clarity by using simple and measurable indicators like vehicle density, traffic flow, and road usage, which makes the results easier to interpret and use for decision-making. Overall, by combining video analytics, data comparison, and adaptive learning, the system offers a practical and scalable solution for modern traffic monitoring.

VIII. CONCLUSION

The real-time traffic monitoring and verification system uses live video feeds along with navigation data to improve the accuracy of traffic information in fast-changing urban environments. By applying deep learning for vehicle detection, it provides clear and reliable real-time insights into road conditions, going beyond what traditional GPS-based or crowd-sourced systems can offer. An important feature of this system is its ability to cross-check video-based traffic data with navigation platform reports, which helps in identifying congestion more accurately, especially in areas where user data is limited or during unexpected traffic situations. With the addition of reinforcement learning, the system can continuously learn from changing traffic patterns and improve its predictions over time. Its flexible and scalable design also allows it to work smoothly with existing smart transportation systems while supporting real-time monitoring and decision-making for both authorities and commuters. In the future, the system can be further improved by including data such as weather updates, road construction information, accident reports, and event schedules to increase accuracy. Using lightweight models and edge computing can help reduce delays and make large-scale deployment more efficient. It can also be expanded to connect multiple cameras across road networks for better city-wide monitoring and vehicle tracking. Adding features like predictive analytics and long-term traffic trend visualization can support better planning and policy decisions. Overall, the system has strong potential to grow into a smart mobility platform that offers dependable real-time insights and supports sustainable, data-driven urban transportation.

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